# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET B

Chapters 6 & 10: **Triangles, Circles** | RBSE Board Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** If $\triangle ABC \sim \triangle QRP$ and $\angle A = 50^\circ$, then $\angle Q$ is: 1

**(a)** $40^\circ$ **(b)** $50^\circ$ **(c)** $130^\circ$ **(d)** $90^\circ$

**2.** In $\triangle ABC$, $DE \parallel BC$ with $AD = 1$ cm, $DB = 2$ cm and $BC = 6$ cm. Then $DE$ is: 1

**(a)** 2 cm **(b)** 1.5 cm **(c)** 3 cm **(d)** 4 cm

**3.** SAS similarity requires: 1

**(a)** all sides proportional
**(b)** a right angle
**(c)** two angles equal
**(d)** one angle equal and the sides including it proportional

**4.** A student states: “Two triangles with equal areas are similar.” The correct response is: 1

**(a)** Correct
**(b)** Incorrect — equal area does not imply equal angles
**(c)** Correct for right triangles
**(d)** Cannot be determined

**5.** $\triangle ABC$ has sides 2, 3, 4 cm and $\triangle DEF$ has sides 4, 6, 8 cm. Student A says they are similar; Student B says they are not. Then: 1

**(a)** Student B is correct **(b)** Student A is correct
**(c)** Both are correct **(d)** Both are wrong

**6.** A line that touches a circle at exactly one point is called a: 1

**(a)** diameter **(b)** secant **(c)** tangent **(d)** chord

**7.** A tangent PQ at P meets OQ where $OQ = 25$ cm and the radius is 7 cm. Then PQ is: 1

**(a)** 32 cm **(b)** 24 cm **(c)** 18 cm **(d)** 20 cm

**8.** Tangents at A and B meet at P. If $\angle AOB = 100^\circ$, then $\angle APB$ is: 1

**(a)** $40^\circ$ **(b)** $50^\circ$ **(c)** $80^\circ$ **(d)** $100^\circ$

**9.** A student states: “Tangents drawn from an external point subtend equal angles at the centre.” The correct response is: 1

**(a)** Correct only for $90^\circ$ **(b)** Correct
**(c)** Incorrect **(d)** Cannot be determined

**10.** A quadrilateral circumscribes a circle with $AB = 5$, $BC = 6$, $CD = 7$ cm. Student A says $AD = 6$ cm; Student B says $AD = 4$ cm. Then: 1

**(a)** Student B is correct **(b)** Student A is correct
**(c)** Both are correct **(d)** Both are wrong

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Are triangles with sides 3, 4, 6 cm and 6, 8, 12 cm similar? State the criterion. 2

**12.** In $\triangle ABC$, $DE \parallel BC$ with $AD = 4x - 3$, $AE = 8x - 7$, $BD = 3x - 1$ and $CE = 5x - 3$. Find $x$. 2

**13.** Tangents PA and PB are drawn from P to a circle. A third tangent touches the circle at C and meets PA at X and PB at Y. If $PA = 10$ cm, find the perimeter of $\triangle PXY$. 2

**14.** A point P is 17 cm from the centre of a circle of radius 8 cm. Find the length of the tangent PT and $\angle OTP$. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** In $\triangle PQR$, $ST \parallel QR$ with S on PQ and T on PR. If $PS = 3$ cm, $SQ = 6$ cm and $PT = 2.5$ cm, find TR. Also find ST if $QR = 9$ cm. 3

**16.** Prove that in two concentric circles, all chords of the larger circle which touch the smaller circle are equal in length. 3

**17.** In $\triangle ABC$, $DE \parallel BC$ (D on AB, E on AC). If $\frac{AD}{DB} = \frac{2}{3}$ and $AC = 10$ cm, find AE and EC. 3

**OR**

The diagonals of trapezium ABCD ($AB \parallel DC$) meet at O, with $AO = 3x - 1$, $OC = 5x - 3$, $BO = 2x + 1$ and $OD = 6x - 5$. Find $x$. 3

**18.** Tangents at A and B to a circle with centre O meet at T. If $\angle ATB = 80^\circ$, find $\angle TAB$ and $\angle OAB$. 3

**OR**

A circle with centre O is inscribed in quadrilateral ABCD with $\angle B = 90^\circ$. If $AD = 23$ cm, $AB = 29$ cm and $DS = 5$ cm (S the point of contact on AD), find the radius. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** CD and GH are respectively the bisectors of $\angle ACB$ and $\angle EGF$, with D on AB and H on FE, of $\triangle ABC \sim \triangle FEG$. Show that 5

(i) $\frac{CD}{GH} = \frac{AC}{FG}$

(ii) $\triangle DCB \sim \triangle HGE$

(iii) $\triangle DCA \sim \triangle HGF$.

**OR**

**Case study:** A mobile tower casts a shadow 30 m long when a 1.5 m tall girl casts a shadow 2 m long. 5

(i) Name the criterion used.

(ii) Find the height of the tower.

(iii) When the girl's shadow is 1.2 m, find the tower's shadow.

**20.** **Case study:** A fire station F is outside a circular lake of radius 120 m with centre O, $FO = 130$ m. Two straight roads FA and FB touch the lake at A and B. 5

(i) Find FA.

(ii) Why is FA = FB?

(iii) Find $\angle OAF$.

(iv) Find the perimeter of OAFB.

**OR**

Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. Using this, if $OP = 5$ cm and a tangent at P meets a line through O at Q with $PQ = 12$ cm, find OQ. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**